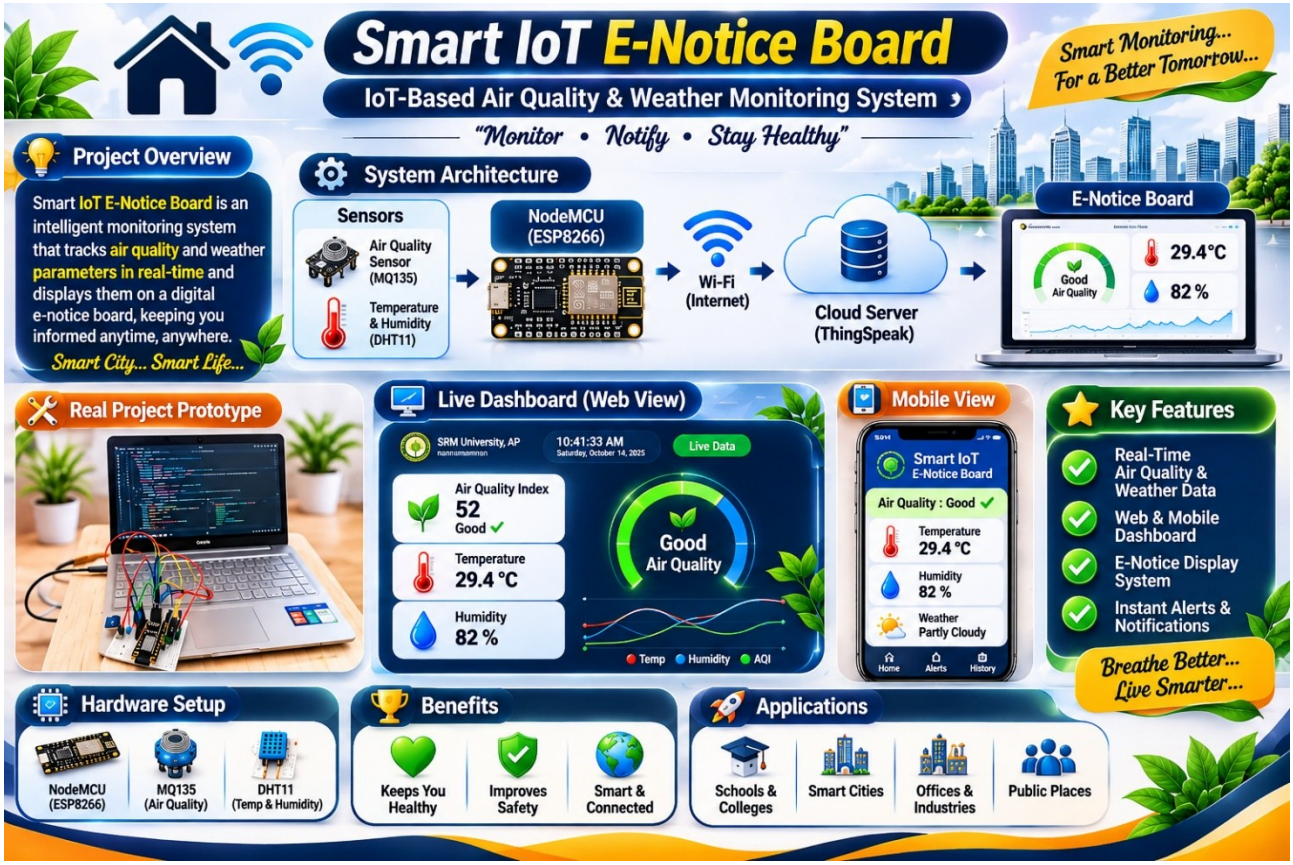
PROJECT DOCUMENTATION

Smart IOT E-Notice Board

IoT-Based Air Quality & Weather Monitoring System



Developed by

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1 Abstract

In this project I built a small Wi-Fi sensor node that keeps an eye on the temperature, humidity and air quality of a room, along with a web page where anyone can see the readings live. The node uses an ESP8266 (NodeMCU) board, a DHT11 sensor for temperature and humidity, and an MQ135 sensor for general air quality. Every ten seconds it sends its readings over a secure HTTPS connection to Supabase, which I use as the online database.

The web page takes those raw numbers and turns them into things people can actually understand: an estimated air quality index, CO_2 level, "feels like" temperature, dew point, absolute humidity and a harmful gas index, plus a chart of the last three hours. The same page also has an E-Notice Board. Authorised staff can post notices with text, images, videos or YouTube links, so the department no longer depends on paper circulars.

Keywords: IoT, ESP8266, NodeMCU, DHT11, MQ135, Supabase, air quality, dashboard, e-notice board.

2 Introduction

2.1 Background

The air we breathe indoors, and the temperature and humidity around us, affect how comfortable and focused we feel. Most campuses never measure any of this. On top of that, notices are still printed and pinned to boards, where they are slow to put up, easy to miss and impossible to search later. Cheap IoT hardware and free cloud tools make it possible to fix both problems together.

2.2 Problem Statement

> **The problem**
> The department has no simple, low-cost way to (a) keep track of room conditions and check them remotely, and (b) send official announcements to students and staff instantly, sorted by category.

2.3 Objectives

- Build a Wi-Fi sensor node that reads temperature, humidity and air quality reliably.
- Send the readings to a cloud database at regular intervals, securely.
- Turn the raw sensor values into numbers and trends that are easy to read.
- Make a dashboard that works on phones and big screens, in light and dark themes.
- Add an E-Notice Board with categories, images and video, and admin controls.
- Keep the whole thing cheap and easy for others to copy and extend.

2.4 Where It Can Be Used

Although this prototype has a single node, the same idea fits many places:

- Classrooms, labs and seminar halls, to check comfort and ventilation.
- Libraries and storage rooms, where temperature and humidity matter.
- Hostel corridors and canteens, to catch smoke or strong fumes early.
- A display screen in the department showing live conditions next to the latest notices.

3 System Overview

3.1 Architecture

The system has three layers:

- **Sensing:** the ESP8266 with the DHT11 and MQ135 collects data from the room.
- **Network and cloud:** Wi-Fi and HTTPS carry each reading to Supabase, which stores it and makes it available to the web app.
- **Application:** a browser-based dashboard and notice board show the live values, the history and the announcements.

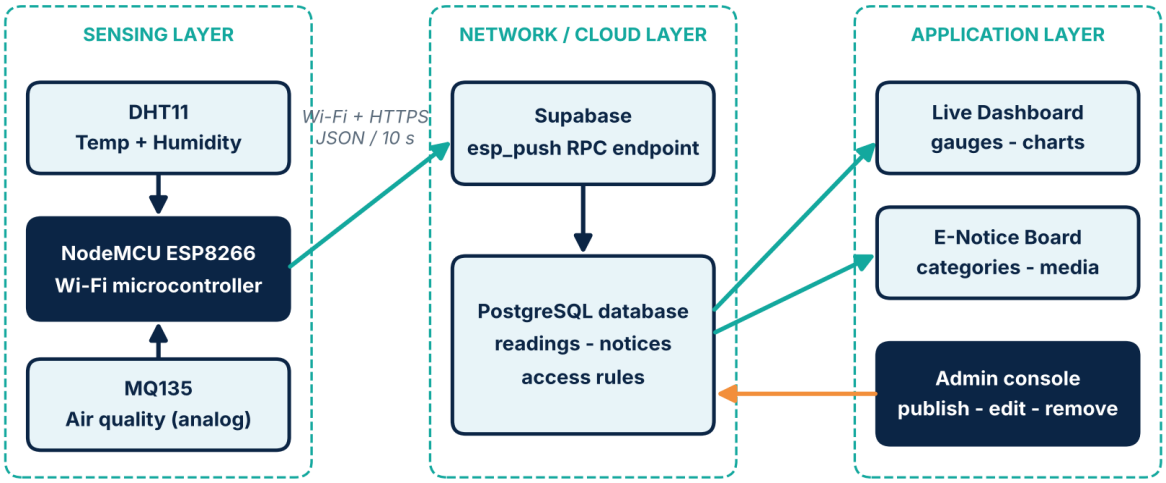


Figure 1: Three-layer system architecture

3.2 How It Works

1. When the board powers up, it joins the campus Wi-Fi and sets up a secure connection to the cloud.
2. Every 10 seconds it reads temperature and humidity from the DHT11 and takes the average of a few readings from the MQ135.
3. It puts the values, the Wi-Fi signal strength and its uptime into a small JSON message and sends it to a protected endpoint in the cloud.
4. The cloud stores the data. The dashboard refreshes and shows the board as online, with the time since the last update.
5. The dashboard works out the extra values (AQI, dew point and so on) and draws the 3-hour chart.
6. An administrator logs in on the notice board section to publish, edit or remove notices.

4 Hardware Description

4.1 Components Used

Component	What it does	Notes
NodeMCU ESP8266 (ESP-12E)	32-bit microcontroller with built-in Wi-Fi; the main controller	Powered through micro-USB from a laptop or adapter
DHT11 sensor module	Measures temperature and relative humidity	Data pin on D4; typical accuracy ±2 °C and ±5 % RH
MQ135 gas sensor module	Analog air-quality sensor (NH_3, NOx, alcohol, benzene, smoke, CO_2)	Output on A0; needs about 2 minutes to warm up
Breadboard	Lets me wire things without soldering	Shared power rails for both sensors
Jumper wires	Male-to-male and male-to-female	Different colours for power, ground and signal
Micro-USB cable	Power and programming	Also used for the serial monitor while testing

4.2 Pin Connections

Sensor pin	NodeMCU pin	Purpose
DHT11 DATA	D4	Digital temperature and humidity data
MQ135 AO	A0	Analog gas level (0 to 3.3 V input range)
DHT11 / MQ135 VCC	3V3 or 5V, depending on the module	Power for the sensors
DHT11 / MQ135 GND	GND	Common ground

> **Please note**
> The analog pin of the NodeMCU can only take 0 to 3.3 V. Some MQ135 modules put out up to 5 V, so in that case a voltage divider (for example 10 kΩ and 20 kΩ) should go between the sensor output and A0, otherwise the board can get damaged.

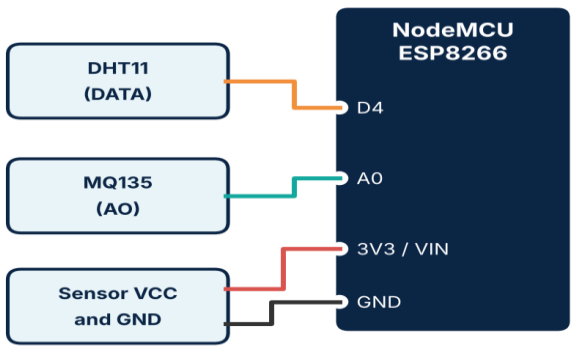


Figure 2: Wiring diagram of the sensor node

4.3 About the Sensors

	DHT11	MQ135
Measures	Temperature and relative humidity	Combined response to NH_3, NOx, alcohol, benzene, smoke and CO_2
Output	Digital, single wire (D4)	Analog voltage read by the 10-bit ADC (0 to 1023)
Accuracy	±2 °C, ±5 % RH (typical)	Only an estimate; should be calibrated in clean air
Start-up	Ready in a second or two	About 2 minutes of warm-up for a stable value
Noise	Bad readings are sent as empty values	Average of 8 samples for each reading

4.4 Circuit and Prototype

These photos show the breadboard prototype and the board connected to my laptop for programming and testing. In the serial monitor you can see each upload succeed, with the server replying HTTP 204 for every reading.

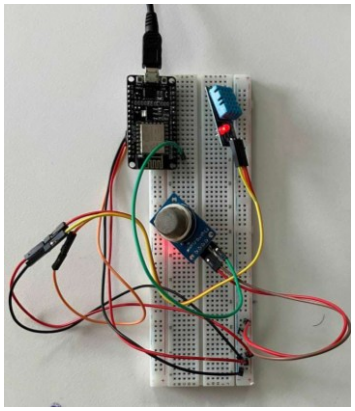


Figure 3: Breadboard circuit with NodeMCU, DHT11 and MQ135

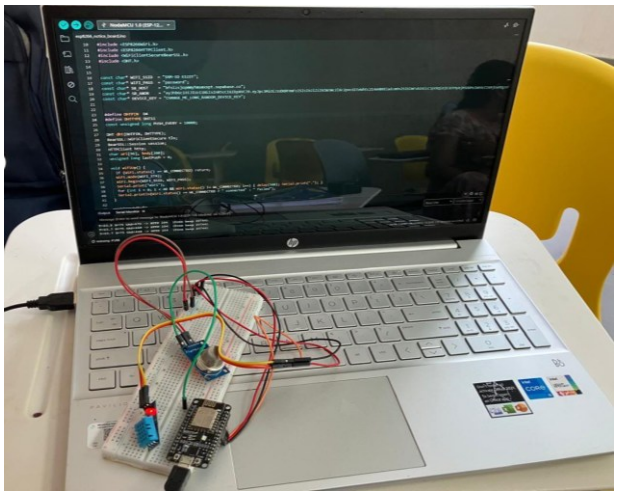


Figure 4: Prototype connected to the laptop, with the Serial Monitor output during testing

5 Software and Cloud Design

5.1 Tools Used

Part	Tool
Firmware	Arduino IDE with the ESP8266 board package (NodeMCU 1.0)
Sensor libraries	DHT sensor library and Adafruit Unified Sensor
Networking	ESP8266 Wi-Fi, HTTP client and BearSSL secure client (HTTPS)
Cloud backend	Supabase (hosted PostgreSQL database with a remote-procedure endpoint)
Frontend	Responsive web app in HTML, CSS and JavaScript, with animated themes

5.2 How the Firmware Works

I kept the firmware small and split it into four jobs:

- **Staying connected:** if the Wi-Fi drops, the board reconnects by itself. I also turned off Wi-Fi power saving because it made the connection steadier.
- **Reading the sensors:** temperature and humidity come from the DHT11. For the MQ135 I take eight quick readings and average them to reduce noise. If the DHT11 gives a bad reading, the board sends an empty value instead of a wrong number.
- **Sending data:** every 10 seconds it sends a JSON message with the device key, temperature, humidity, raw gas value, Wi-Fi signal strength (RSSI) and uptime. It reuses one secure connection and caches the TLS session, so it does not repeat the heavy handshake every time. This keeps memory use low.
- **Recovering from errors:** if a send fails, the board drops the broken connection and tries again after about 3 seconds. It also prints the reset reason and free memory on the serial monitor, which helped a lot while debugging.

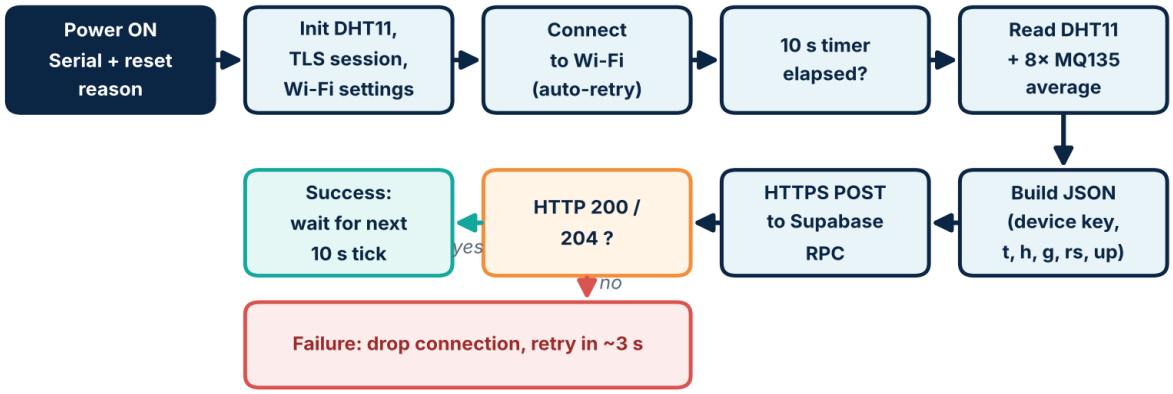


Figure 5: Firmware flow, from power-on to the repeating cycle

5.3 Data Sent by the Node

Field	Meaning
device_key	Secret key that tells the backend this is the authorised node
t	Temperature in °C (one decimal place)
h	Relative humidity in %
g	Averaged raw analog value from the MQ135
rs	Wi-Fi signal strength (RSSI) in dBm
up	How long the node has been running, in seconds

5.4 Cloud Backend

Supabase gives me the database and a protected endpoint (esp_push) that the node calls over HTTPS. The endpoint checks the device key first, so only my node can write sensor data. The web app then reads the stored values to show the live readings and the three-hour history.

- **Online status:** the board counts as online while heartbeats keep arriving, and the dashboard shows how long ago the last one came.

- **Notices only when online:** notices can be read and edited only while the board is online (a heartbeat in the last 60 seconds). This rule is enforced in the database itself, so it cannot be skipped by editing the web page.
- **Admin login:** publishing, editing and removing notices needs an administrator login.

5.5 Values Shown on the Dashboard

The MQ135 gives only one analog number, so the dashboard estimates the air-quality values from it. The comfort values are calculated from temperature and humidity.

Value	What it means
AQI (estimated)	Air quality index with a label such as Satisfactory
Carbon dioxide	Estimated CO_2 in ppm, with a rating such as Good
Feels like	How warm it actually feels, combining temperature and humidity
Dew point	Temperature at which moisture starts to condense, with a comfort label such as Oppressive
Absolute humidity	Amount of water vapour in the air, in g/m^3
Harmful gas index	Combined indication of gases like NH_3, benzene and smoke, as a percentage
Gas breakdown	Separate estimates for CO_2, NH_3, NOx, alcohol vapour, benzene, smoke, CO and toluene, each with a Low / Moderate / High bar

6 Web Application

6.1 Live Dashboard

At the top of the dashboard there is the SRM University AP name, a live clock and date, and a status chip that shows whether the board is online and how long ago it last sent data. A small credits panel mentions the developer.

The left card, Air Quality (MQ135), has three pages that you can swipe through. Page 1 is a gauge with the current value and a status like "Normal". Page 2 lists AQI, CO_2, feels-like, dew point, absolute humidity and the harmful gas index. Page 3 shows each gas separately. The right card shows the current temperature and humidity and a three-hour chart, with tabs to switch between Temperature, CO_2 and AQI. A line under the chart gives the lowest and highest values in that period.

6.2 Themes

I wanted the page to look good on a display wall as well as on a phone, so it has an automatic theme, a few animated themes (sunrise, autumn leaves, rain and snow) and a light/dark toggle. The screenshots below show the same live data in different themes.



Figure 6: Dark autumn theme showing AQI, CO_2, feels-like, dew point, absolute humidity and harmful gas index

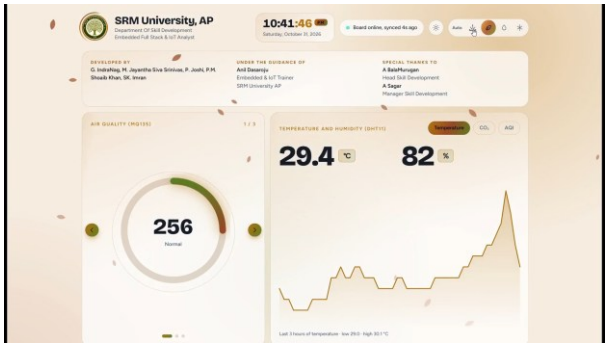


Figure 7: Light theme, air-quality gauge (page 1 of 3) and temperature/humidity chart

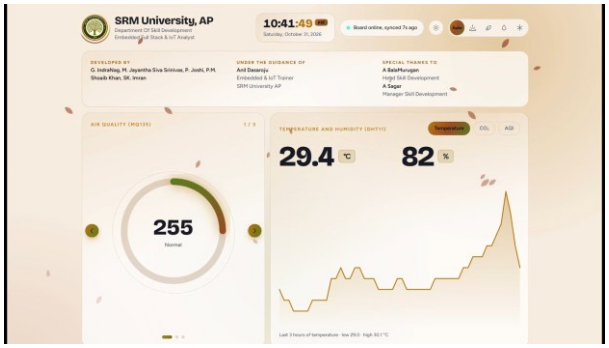


Figure 8: Light theme, gauge updated on the next refresh



Figure 9: Dark rain theme, per-gas breakdown (page 3 of 3)

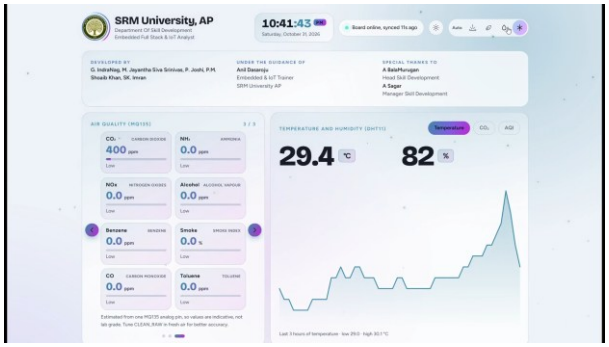


Figure 10: Light snow theme, gas breakdown and temperature chart

6.3 E-Notice Board

Below the live data is the E-Notice Board, which replaces paper circulars. Notices can be filtered by category: Holiday Circular, Examination, Academic, Placements, Event, Special Day and General.

After logging in, an administrator picks a category, types a title and message, optionally adds an image, a video or a YouTube link, and presses Publish (or Clear to start over). Published notices appear in a feed with their category tag, a NEW marker for recent posts, the date and time, and Edit and Remove buttons. Logout ends the admin session.

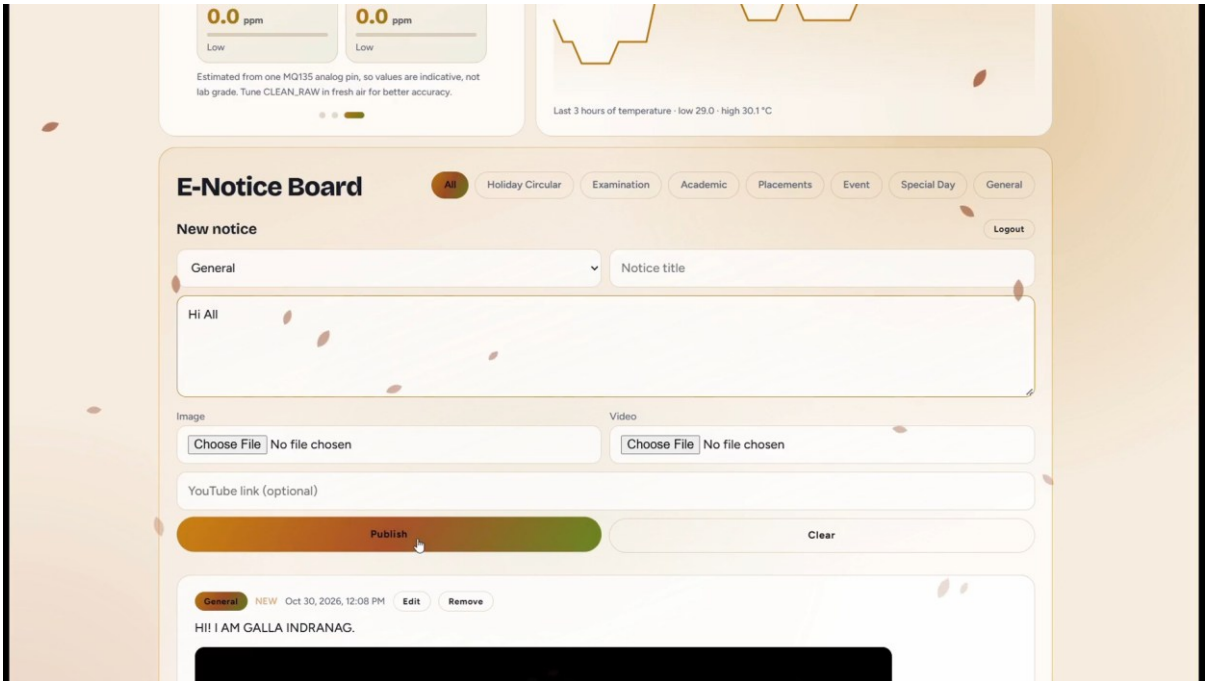


Figure 11: E-Notice Board with category filters, the notice form and the published feed

7 How to Set It Up

7.1 What You Need

- A NodeMCU ESP8266, a DHT11 module, an MQ135 module, a breadboard and jumper wires.
- Arduino IDE with the ESP8266 board package, plus the DHT sensor library and Adafruit Unified Sensor library.
- A Supabase project with the readings table, the notices table and the esp_push function.
- A Wi-Fi network that lets the node reach the internet over HTTPS.

7.2 Steps

1. Wire the sensors as shown in Section 4.2. Add the voltage divider if your MQ135 can output 5 V.
2. Install the libraries and select the NodeMCU 1.0 board in the Arduino IDE.
3. In the firmware, enter your Wi-Fi name and password, the Supabase project address and public key, and a long random device key.
4. Add the same device key in Supabase so the esp_push function accepts your node.
5. Upload the firmware, open the Serial Monitor at 115200 baud and wait for "HTTP 204" after each reading.
6. Give the MQ135 about two minutes to warm up, preferably in clean air, before trusting its values.
7. Open the web app, check that the board shows as online, then log in as admin and try publishing a test notice.

8 Testing and Results

Test	What I saw	Result
Serial monitor upload	Temperature, humidity and gas values printed every cycle, followed by "HTTP 204" (for example about 23.8 °C, 75 % RH and a raw gas value near 470 in the lab)	Pass
Real-time delivery	The dashboard showed the board online, with the last sync only a few seconds old	Pass
Live reading	29.4 °C and 82 % humidity on screen, with a three-hour trend between about 29.0 °C and 30.1 °C	Pass
Memory stability	Free heap stayed around 20 KB over repeated sends, so reusing the connection works	Pass
Dashboard features	All themes, the carousel and the chart tabs worked	Pass
Notice board	Notices could be published, shown, edited and removed	Pass

9 Troubleshooting

Problem	Likely cause	What to try
Serial monitor says Wi-Fi failed	Wrong name or password, or weak signal	Check the credentials and move closer to the router
HTTP code is negative	Broken secure connection or no internet	The firmware retries in 3 seconds; check the network
HTTP 401 or 403, or an error message	Wrong public key, or device key not registered	Check both keys and the esp_push access rules
Temperature or humidity is empty	DHT11 wiring or timing problem	Check D4 and the power; give it a moment after start-up
Gas value stuck near 0 or 1023	MQ135 not warmed up, or wired wrongly	Wait 2 minutes; check A0 and the divider
Board shows offline	No heartbeat for over 60 seconds	Check power and Wi-Fi; read the reset reason in the serial monitor
Board keeps restarting	Weak USB power or low memory	Use a good adapter and cable; watch the free heap

10 Security and Limitations

10.1 Security

- **Encrypted connection:** every reading goes over HTTPS.
- **Device key:** the backend accepts data only when the correct key is sent.

- **Database rules:** access to notices is controlled inside the database, and admin actions need a login.

10.2 Limitations

- The DHT11 is not very accurate. Sensors like the DHT22 or BME280 would be better.
- All gas values are estimated from a single MQ135 analog pin, so they are only indicative, not lab grade. The sensor needs warm-up time and calibration in clean air.
- To keep things simple, the prototype does not check the server certificate on the secure connection. For real deployment, certificate pinning should be added.
- The Wi-Fi details and device key are stored in the firmware. They should be changed from the defaults and kept private.
- Only one sensor node is supported right now.

11 Future Scope

- Use a more accurate temperature sensor and add sensors for dust (PM2.5/PM10) and CO_2.
- Support several nodes and rooms, with a location selector on the dashboard.
- Send alerts by email, SMS or push notification when air quality or temperature goes beyond safe limits.
- Allow data export and weekly or monthly reports.
- Add over-the-air (OTA) firmware updates, and a battery or solar option.
- Send notice board announcements to students as mobile notifications.
- Make an enclosure and a proper PCB so the node can be installed permanently.

12 Conclusion

This project set out to answer a simple question: can a department monitor the air in its rooms and share official notices digitally without spending much money? After building and testing the system, the answer is yes. With a NodeMCU ESP8266, a DHT11 sensor, an MQ135 gas sensor and a free Supabase account, I was able to put together a working system that measures, stores, shows and announces, all from one small board and one web page.

The sensor node does its job quietly and reliably. It reads the sensors every ten seconds, sends the data over an encrypted connection, reuses the same secure link to save memory, and recovers by itself when the Wi-Fi or the connection drops. During testing the serial monitor returned HTTP 204 for every reading, the free memory stayed steady at around 20 KB, and the dashboard showed the board as online with the last update only a few seconds old.

The web application turns raw numbers into information people can understand. Instead of a bare analog value, the user sees an estimated air quality index, CO_2 level, feels-like temperature, dew point and a gas-by-gas breakdown, along with a three-hour history chart. The themes and light/dark mode make it comfortable to use on a phone or on a large display wall. The E-Notice Board adds a second purpose to the same system. Notices are sorted by category, can include images, videos and YouTube links, and can be published, edited or removed by an administrator in a few seconds, which is far faster than printing and pinning a circular.

Building this project also taught me a lot beyond the circuit itself. I learned how to connect a microcontroller to a cloud database securely, how to keep an ESP8266 stable with limited memory, how to design database rules so that access is controlled on the server and not just in the browser, and how to present sensor data in a way that non-technical users can follow. Debugging the connection problems, using the reset reason and free heap values on the serial monitor, was one of the most useful parts of the process.

The project does have limits. The DHT11 is not very accurate, the gas values are estimates from a single MQ135 pin, the prototype supports only one node, and the secure connection does not yet verify the server certificate. These are known

and fixable, and they are listed in this document so that anyone building on this work knows where to improve it. Better sensors, calibration in clean air, certificate pinning, multiple nodes, alerts and over-the-air updates are the natural next steps.

In summary, the system meets all the objectives set at the start: a reliable Wi-Fi sensor node, secure data transfer, readable metrics and trends, a responsive themed dashboard, and a practical E-Notice Board, all at a very low cost. Because the design is simple and the parts are easy to find, it can be copied by other students and departments and scaled up to cover classrooms, labs, hostels and eventually the whole campus.

A Glossary

Term	Meaning
IoT	Internet of Things: everyday devices connected to the internet to send or receive data
ESP8266 / NodeMCU	A low-cost Wi-Fi microcontroller and its popular development board
AQI	Air Quality Index, one number that summarises how clean or polluted the air is
RSSI	Received Signal Strength Indicator, the Wi-Fi signal strength in dBm
TLS / HTTPS	Encryption that protects data travelling between the node and the cloud
RPC	Remote procedure call, a named function in the backend that the node can trigger
Heartbeat	The regular 10-second message that tells the cloud the node is alive
Heap	The free working memory of the microcontroller
Dew point	The temperature at which water vapour starts to turn into liquid

13 Video

https://drive.google.com/file/d/1i5qQXlPrp4aliGqkhX3sRe1NY1DVjebp/view?usp=sharing

14 Website Link

https://srm-board.onrender.com/